Self-Justification Ex Machina

jpt4*

In memory of Dan E. Willard (1948–2023)

Abstract

Sebastian Oberhoff's *Incompleteness Ex Machina* proves Gödel's incompleteness theorems twice, once with sentences and once with programs. His proof of the second theorem has three steps, and he takes one of them "with some faith". We argue that this step is a part of Gödel's work in its own right, that it has a price, measured in bits of proof, and that the price list is a recipe for machines that know they are consistent. A consistent system that proves every true Σ_1 sentence cannot prove that Oberhoff's program $\mathsf{P}(\mathsf{P})$ never halts. His three steps together turn the system's consistency into exactly that claim, so a system that proves its own consistency must fail one of them. Our witness is Dan Willard's 2005 self-justifying axiom system $\mathrm{IS}_D(A)$, with semantic tableaux or tableau lists as its deduction method D. Every Π_1 theorem of its base theory A is true, and A proves every Π_1 theorem of Peano arithmetic in Willard's language. The system proves every true Σ_1 sentence, it obeys the first incompleteness theorem, it proves its own consistency, and, by Willard's theorem, it is consistent. Two of the three steps are theorems of it, and the faith step is not. At the Gödel sentence, built with the conventions we fix, the faith step is provably equivalent to the Gödel sentence itself. In its uniform form it fails because the system cannot prove that every number can be written down. We read Willard's construction as four design rules, each backed by a theorem. A program running on the resulting machine gains a certified guarantee that no two certificates the machine accepts contradict each other, which no consistent, recursively axiomatized extension of $I\Sigma_1$ can prove of its own checker. It does not gain the reflection principle, which at the Gödel sentence would need the faith step as well. Granted, as an axiom, that one large enough number exists, Willard's tableau-list system over Peano arithmetic in his language becomes inconsistent (his Theorem 6). The printed proof of that theorem has a gap, for which we sketch a repair.

Another way to motivate the axiom system IS is to consider a philosopher's response to the question of whether multiplication is a function. His first answer would likely be "of course yes". But if one then asked the philosopher to construct a sequence of natural numbers $a_0, a_1, a_2, a_3, \ldots$ where $a_0 \geq 2$ and $a_{i+1} = (a_i)^2$, a hedge would quickly follow because a_n requires at least 2^n bits. The expected response would then be that "multiplication is a function in theory but not always in practice". — Dan Willard, SUNY Albany TR 93-10 (1994)

1 One Step On Faith

Oberhoff's essay [Oberhoff 2018] writes out Gödel's first incompleteness theorem, Rosser's improvement of it and the second incompleteness theorem as explicit case analyses. Then it replaces Gödel's sentence G ("I am not provable") by $\ddot{\mathsf{G}} =$ «$\mathsf{P}(\mathsf{P})$ doesn't halt», where P is the program

*Revision of 27 October 2026, prepared with Claude (Anthropic) in the `sjas` research repository, `https://github.com/jpt4/sjas`. The pages and proof statuses of Willard's results are those recorded in the repository's codification of his papers (`nachlass/codification`); Appendix D lists the ones used here.

```
P(A) {
  for s ∈ all possible strings {
    if s is a proof in 𝓕 that A(A) doesn't halt {
      halt;
    }
  }
}
```

and observes that each case analysis can be copied almost word for word. The punchline is that $\ddot{\mathsf{G}}$ is G wearing Clark Kent's glasses: $\mathsf{P}(\mathsf{P})$ is searching $\mathcal{F}$'s proofs for a proof of $\ddot{\mathsf{G}}$, so $\ddot{\mathsf{G}}$ says "I am not provable". The moral is a refactoring. The hard part of Gödel's work was teaching arithmetic to compute, which Oberhoff packages as

> **Theorem 7** (Turing Completeness of Elementary Arithmetic). *Every formal system capable of reasoning about elementary arithmetic can also reason about algorithms,*

and leaves "to smarter people". The incompleteness theorems themselves are victory laps.

For the first incompleteness theorem we agree completely. This essay is about one step of the second, the only step that Oberhoff marks as taken on trust:

> Now we simply observe (with some faith) that there wasn't actually anything in the reasoning we used to prove this that we couldn't carry out just as well in $\mathcal{F}$.

For Peano arithmetic and its relatives the faith is well placed. But some formal systems reason about algorithms in every sense the first theorem needs, and for them it is misplaced. Their proofs are cut-free, or reuse only simple lemmas. They prove their own consistency, and they are consistent, provided only that their base theory tells no lies about universal statements. Dan Willard built them, in papers from 1993 to 2020 [Willard TR 93-10, Willard 2001, Willard 2005, Willard 2006a, Willard 2020]. This essay tells their story in Oberhoff's language. Our headline is that the faith step is not part of the victory lap. It is a third part of Gödel's work, and it has a price, which Section 5 measures in bits of proof. The price list is also a recipe. Section 9 reads Willard's construction as instructions for building a machine that knows it is consistent, and Section 10 says what a program gains by running on one, and what it does not.

Ground rules. Oberhoff's footnote 2 lets "=" relate sentences that are merely logically equivalent, "provided this is obvious". We cannot afford that luxury. Willard's systems are self-justifying for one deduction method and inconsistent for another, and consistent under one notion of consistency but not the next. The proofs that tell these cases apart measure Gödel numbers in bits. A system may prove two sentences equivalent without proving that one is provable exactly when the other is. So a formal system, here, is always an axiom system $\mathcal{F}$ together with a deduction method D. Every method we use is *sound and complete*: $\mathcal{F}$ proves φ by D exactly when φ holds in every model of $\mathcal{F}$. Proof length is what separates them, and self-justification is a phenomenon of length. Here is the fact we use most:

Fact A (Closure; Willard 2001, Lemma 7.1B)**.** *If D is sound and complete for $\mathcal{F}$, $\mathcal{F}$ proves φ and ψ, and $\varphi \wedge \psi \supset \chi$ holds in every model of $\mathcal{F}$, then $\mathcal{F}$ proves χ.*

Everything else is fixed in Appendix A: the languages, the Gödel numbering, how a negation is put into prenex form, and which sentence the consistency axiom counts as another's negation. The proofs that depend on these choices are there too. Every consistency result below, Willard's and

ours, is a theorem of ordinary mathematics under a hypothesis on a base theory, and none shows a system consistent outright. We cite Willard's results by year and item, as in "2005, Theorem 3". Appendix D gives their pages and proof statuses. A dagger, as in "2004, Theorem $2^{\dagger}$", marks a result printed without a complete proof (with a sketch, a citation, or no proof at all), or one whose printed proof we find incomplete at our level of detail. Results marked *ours* are proved here.

2 Watching And Knowing

Here is the first half of Oberhoff's proof that $\ddot{\mathsf{G}}$ is undecidable:

Suppose $\ddot{\mathsf{G}}$ is provable:
Or $\neg\ddot{\mathsf{G}}$ isn't provable:

⇒ Because $\ddot{\mathsf{G}}$ is provable $\mathsf{P}(\mathsf{P})$ will eventually find such a proof after some sequence of steps which we can write out in $\mathcal{F}$.

⇒ Because this sequence leads to a terminal state this gives a proof of «$\mathsf{P}(\mathsf{P})$ halts» $= \neg\ddot{\mathsf{G}}$. ↯

His Theorem 6 asks $\mathcal{F}$ to run this argument on itself. Write Prov for provability in $\mathcal{F}$, and H for the Σ_1 sentence «$\mathsf{P}(\mathsf{P})$ halts», in the form that Appendix A fixes.[1] As statements that $\mathcal{F}$ must prove, the argument has three steps.

(1) $\mathsf{H} \supset \mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$: «if $\mathsf{P}(\mathsf{P})$ halts, then $\ddot{\mathsf{G}}$ is provable».

(2) $\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner) \supset \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner)$: «if $\ddot{\mathsf{G}}$ is provable, then so is «$\mathsf{P}(\mathsf{P})$ halts»».

(3) $\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner) \wedge \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner) \supset \neg \mathit{Con}$: «then $\mathcal{F}$ is inconsistent».

Chaining them gives $\mathcal{F} \vdash \mathit{Con} \supset \ddot{\mathsf{G}}$, which is what Theorem 6 needs. We call this implication the *bridge*. It leads from $\mathcal{F}$'s consistency to the Gödel sentence, and the three steps are its parts. Step (1) is the fixed point. $\mathcal{F}$ proves it once it proves one true bounded sentence and that $\leq$ is trichotomous at one number (Fact N). Step (3) comes free when *Con* says that no sentence and its negation both have proofs. If it says only that $0 = 1$ has no proof, step (3) is a formalized combination of two proofs into one, and for cut-free deduction the combined proof can be far longer than the two (Section 7). Step (2) is "$\mathsf{P}(\mathsf{P})$ will eventually find such a proof after some sequence of steps which we can write out in $\mathcal{F}$". Inside $\mathcal{F}$ it reads: *for every proof s of $\ddot{\mathsf{G}}$ there is a proof that $\mathsf{P}(\mathsf{P})$ halts*, namely the one that exhibits s and checks it.

Here, in advance, is where the three steps stand. The first row is the systems of Proposition C, Peano arithmetic among them. The second is Willard's 2005 system $\mathrm{IS}_D(A)$ of Section 4, under the hypotheses of Corollary J.

	Step (1)	Step (2)	Step (3)
Extensions of $I\Sigma_1$	theorem	theorem	theorem
$\mathrm{IS}_D(A)$	theorem	**not a theorem**	theorem

In $\mathrm{IS}_D(A)$, step (1) follows from the fixed point, and step (3) is its consistency axiom read at $\ddot{\mathsf{G}}$ and its negation. So exactly one step fails, the faith step, and it fails in two forms. Its uniform form fails for reasons of growth (Section 5). At $\ddot{\mathsf{G}}$ itself it is provably equivalent to $\ddot{\mathsf{G}}$, and so it is

[1] That form has exactly the proofs of $\neg\ddot{\mathsf{G}}$, and the consistency axiom counts it as $\ddot{\mathsf{G}}$'s negation. Another Σ_1 rendering of «$\mathsf{P}(\mathsf{P})$ halts» would be a different sentence, with different proofs.

unprovable for the reason $\ddot{\mathsf{G}}$ is (Section 6). For Willard's 2001 system, step (2) and the Level(0−) form of step (3) are not both theorems, and we do not know which one fails (Appendix B).

In the first incompleteness theorem, "which we can write out in $\mathcal{F}$" was a fact *about* $\mathcal{F}$, established by us from outside: we are handed a halting run and we produce the proof. In the second it has to be a theorem *of* $\mathcal{F}$, about every s at once. The two deserve separate names.

Definition (Watching)**.** $\mathcal{F}$ *watches* if it proves every true Σ_1 sentence.

Definition (Knowing that one watches)**.** $\mathcal{F}$ *knows that it watches* at a sentence σ if $\mathcal{F} \vdash \sigma \supset \mathrm{Prov}(\ulcorner\sigma\urcorner)$. It knows that it watches if it does so at every Σ_1 sentence.

A program halts exactly when a Σ_1 sentence about it is true. So watching is Σ_1-completeness, and knowing that one watches is its formalized version, taken one sentence at a time. Section 5 adds the uniform version. The third Hilbert–Bernays–Löb derivability condition [Hilbert–Bernays 1939, Löb 1955], $\mathrm{Prov}(\ulcorner\varphi\urcorner) \supset \mathrm{Prov}(\ulcorner\mathrm{Prov}(\ulcorner\varphi\urcorner)\urcorner)$, is knowing that one watches at the halting statements of proof searches.

Lemma B (ours)**.** *Let $\mathcal{F}$ watch, with a sound and complete deduction method and a correct proof predicate.*

(a) $\mathcal{F}$ knows that it watches at every true Σ_1 *sentence.*

(b) If $\mathcal{F}$ proves the fixed point, then step (2) is $\mathcal{F}$-provably equivalent to knowing that it watches at H*, that is, to* $\mathsf{H} \supset \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner)$.

Proof. (a) If σ is true, $\mathcal{F}$ proves it by watching. So $\mathrm{Prov}(\ulcorner\sigma\urcorner)$ is a true Σ_1 sentence, $\mathcal{F}$ proves that too, and $\sigma \supset \mathrm{Prov}(\ulcorner\sigma\urcorner)$ follows by Fact A. (b) H is logically equivalent to $\neg\ddot{\mathsf{G}}$, and the fixed point makes $\neg\ddot{\mathsf{G}}$ equivalent to $\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$, so $\mathcal{F}$ proves $\mathsf{H} \leftrightarrow \mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$. The two implications differ only by this replacement in the antecedent, so $\mathcal{F}$ proves them equivalent (Fact A). □

So the faith step is not a claim about runs that happen. Those a watching system sees, and Lemma B(a) gives it the knowledge for free. The faith is a claim about a run that does not happen. If $\ddot{\mathsf{G}}$ is true, then H is false, and step (2) says that *had* $\mathsf{P}(\mathsf{P})$ halted, $\mathcal{F}$ would have seen it.

Every other result in Oberhoff's essay uses at most watching and honesty. Rosser's version also needs one Π_1 fact: a run stuck in `while true {}` stays stuck. As his footnote 12 says, a system that cannot prove this "ought to be ashamed of itself". The classical proof of the second theorem runs through the derivability conditions. The computational retelling shows what the third one says.

3 A Very Stable Genius

Oberhoff's moral for the second theorem is "if you declare yourself a very stable genius, you're not". Let us build the genius with his tools. Fix a base system $\mathcal{F}_0$ with a deduction method, and consider

```
C(A) {
  for s ∈ all possible strings {
    if s is a contradiction in F_0 + «A(A) doesn't halt» {
      halt;
    }
  }
}
```

where a *contradiction* is either a proof of $0 = 1$ or proofs of a Π_1 sentence and of its negation.

Willard calls the two corresponding notions of consistency *Level(0−)* and *Level(1)*.[2] Let

$$\ddot{\mathsf{W}} = \text{«}\mathsf{C}(\mathsf{C}) \text{ doesn't halt»}, \qquad \mathcal{F} = \mathcal{F}_0 + \ddot{\mathsf{W}}.$$

$\mathsf{C}(\mathsf{C})$ is C run on its own text. So it searches for a contradiction in $\mathcal{F}_0 + \text{«}\mathsf{C}(\mathsf{C}) \text{ doesn't halt»}$, which is $\mathcal{F}$, and it halts exactly when $\mathcal{F}$ is inconsistent. Just as $\ddot{\mathsf{G}}$ says "I am not provable", $\ddot{\mathsf{W}}$ says "$\mathcal{F}$ is consistent", and $\ddot{\mathsf{W}}$ is one of $\mathcal{F}$'s own axioms. So $\mathcal{F}$ proves its own consistency in one line.

Proposition C (The genius falls; standard)**.** *Let $\mathcal{F}_0$ be a recursively axiomatized extension of $I\Sigma_1$, with Hilbert-style deduction and the canonical provability predicate. Then, at either level, $\mathcal{F} = \mathcal{F}_0 + \ddot{\mathsf{W}}$ is inconsistent and $\mathsf{C}(\mathsf{C})$ halts.*

Proof. Suppose $\mathcal{F}$ is consistent.

- ⇒ $\mathcal{F}$ proves its own consistency. $\ddot{\mathsf{W}}$ is an axiom, and by the diagonal lemma $I\Sigma_1$ proves $\ddot{\mathsf{W}}$ equivalent to $\mathcal{F}$'s consistency. It also proves the two levels equivalent: explosion turns a proof of $0 = 1$ into proofs of everything, and modus ponens turns proofs of a sentence and its negation into a proof of $0 = 1$.
- ⇒ The derivability conditions hold for $\mathcal{F}$, provably in $I\Sigma_1$ [Hájek–Pudlák 1993]. In our terms $\mathcal{F}$ knows that it watches, and steps (1)–(3) are theorems of $\mathcal{F}$.
- ⇒ So Oberhoff's Theorem 6 applies, with its faith step now a theorem, and $\mathcal{F}$ cannot prove its own consistency. ↯

So $\mathcal{F}$ is inconsistent. A contradiction in $\mathcal{F}$ is then a finite object, and $\mathsf{C}(\mathsf{C})$ finds it. □

Willard records the old verdict on such systems: "Kleene, Rogers and Jeroslow [...] each warned their readers that most axiom systems, similar to α^*, were useless on account of their inconsistency, although they were technically well-defined" (2005, §1). Constructing $\ddot{\mathsf{W}}$, he adds, is "relatively easy, via an application of the Fixed Point Theorem, but this sentence is ironically typically useless!" (2016, Example 3.5). To make it useful, he changed everything else.

4 Willard's Machine

Willard keeps C and $\ddot{\mathsf{W}}$, and changes the language, the deduction method and the base. We describe his 2005 system $\mathrm{IS}_D(A)$. His 2001 system, which asserts only Level(0−) consistency, is in Appendix B.

The language. There are constants for 0 and 1, two growth functions, + and $\mathit{Double}(x) = x + x$, and six functions that never exceed their largest argument: truncated subtraction, division, roots, maximum, logarithm and a bit count. Every number $n \geq 2$ has a name $\bar{n}$ with at most $2 \log_2 n$ growth symbols, which is its binary notation. Multiplication is not a function symbol. Its graph is a bounded formula, so the system knows the laws of multiplication but not that every product exists. "Bounded", "Π_1" and "Σ_1" mean what they usually mean, for these function symbols. Peano arithmetic in this language is called PA+.

[2] We picked C for consistency, and because it is a circle that never quite closes. Section 4 explains the joke.

The axioms. A is any axiom system in this language with a bounded proof predicate, and D is a deduction method. The axioms of $\mathrm{IS}_D(A)$ fall into four groups (2005, §3).

- **Group-0** (names) defines 0, 1, + and *Double*.
- **Group-1** (watching) is a finite set of true Π_1 sentences that proves every true bounded sentence.
- **Group-2** (inheritance) is $\forall p\,[\mathrm{Prf}_A(\ulcorner\Phi\urcorner, p) \supset \Phi]$, for every Π_1 sentence Φ. Whatever Π_1 sentence A proves, $\mathrm{IS}_D(A)$ proves too.
- **Group-3** (self-reference) is $\ddot{\mathsf{W}}$ at Level(1): no Π_1 sentence and its negation both have D-proofs from $\mathrm{IS}_D(A)$. It is built by the same fixed point as $\ddot{\mathsf{G}}$ (Appendix A).

The deduction methods. A *semantic tableau* proof of Φ is a tree whose root is $\neg\Phi$ in prenex form. It grows by elimination rules alone, with no cut, and every branch closes with some sentence and its negation. A *tableau list* is a list of tableau proofs, each of which may use the earlier conclusions as axioms, all of them Π_1 or Σ_1. This is modus ponens restricted to lemmas with one unbounded quantifier block.

Theorem D (Willard 2005, Theorems 3 and 5)**.** *If every* Π_1 *theorem of* A *is true, then* $\mathrm{IS}_D(A)$ *is consistent, both for semantic tableaux and for tableau lists. With the conventions of Appendix A, its Group-3 axiom is in fact true.*

With Hilbert-style deduction the same construction "will be automatically inconsistent", by theorems of Pudlák and Solovay (2005, §3) that Section 7 returns to. Meanwhile $\mathrm{IS}_D(A)$ reasons about algorithms perfectly well.

Lemma E (ours)**.** *For either deduction method,* $\mathrm{IS}_D(A)$ *watches. If every* Π_1 *theorem of* A *is true, then every theorem of* $\mathrm{IS}_D(A)$ *is true, and the proof predicate that its Group-3 axiom uses, Willard's SubstPrf*$(\bar{n}_3, \cdot, \cdot)$ *(Appendix A), is a correct bounded proof predicate for* $\mathrm{IS}_D(A)$*.*

Proof. A true Σ_1 sentence $\exists\vec{x}\,\delta(\vec{x})$ with witnesses $\vec{k}$ follows by logic from the true bounded sentence $\delta(\bar{\vec{k}})$, which Groups 0 and 1 prove, and Fact A does the rest. Truth and correctness are proved in Appendix A. □

Suppose, then, that every Π_1 theorem of A is true and that A proves every Π_1 theorem of PA+. Then $\mathrm{IS}_D(A)$ obeys the first incompleteness theorem, in Rosser's form as well. $\ddot{\mathsf{G}}$ is unprovable by Corollary J, $\neg\ddot{\mathsf{G}}$ is false and so unprovable, and Rosser's loop fact is inherited from PA+ through Group-2 (Appendix A). In 1994 Willard described his first system in these terms, as "a class of structures, lying halfway between Peano Arithmetic and Presburger Arithmetic, which satisfy Godel's First but not Second Incompleteness Theorem" (TR 93-10).

Why the genius survives

Willard's 2005 consistency proof rests on one idea, *horizons*. Call a universal sentence $\forall\vec{x}\,\psi$ *true up to* a if ψ holds whenever every $x_i \le a$. Call an existential sentence $\exists\vec{x}\,\psi$ *witnessed below* b if some $\vec{x} \le b$ satisfies ψ. Bounded quantifiers inside ψ are left alone. For $a > b \ge 1$, a system is *Normed*(a, b) if each axiom is a universal sentence true up to a or an existential sentence witnessed below b (Definition 4). A deduction method is *θ-compactified* if proofs cannot see much past their axioms. Suppose $p \le \lceil a/b \rceil + 1$ is a proof of Φ from a Normed(a, b) system. Then an existential Φ

must be witnessed below $b\,p^{\theta}$, and a universal Φ must be true up to a/p^{θ} (Definition 5). Tableaux and tableau lists are $\frac{1}{4}$-compactified (Theorems 2 and 4). Here is Theorem 1 with $\theta = \frac{1}{4}$ (Figure 1).

Suppose $\mathrm{IS}_D(A)$ **is inconsistent:**

$\Rightarrow$ Then $\mathsf{C}(\mathsf{C})$ halts. There are proofs j_1 of a Π_1 sentence Φ and j_2 of its negation, and by Appendix A j_2 also proves the Σ_1 sentence Ψ that negates Φ. Choose such a pair with $m = \max(j_1, j_2)$ as small as possible.

$\Rightarrow$ Groups 0–2 are true. Group-3 is false, but its least counterexample is this pair, so it is true up to $m - 1$. Hence $\mathrm{IS}_D(A)$ is Normed$(m - 1, 1)$.

$\Rightarrow$ So j_1 certifies Φ true up to $(m-1)/m^{1/4}$, and j_2 finds a witness for Ψ below $m^{1/4}$.

$\Rightarrow$ But $(m-1)/m^{1/4} > m^{1/4}$, so the witness for Ψ lies where Φ has been certified.[3] ↯

The work is in the third arrow, and it is a program. Suppose a tableau p proves an existential sentence Φ from a Normed(a, b) system, but Φ has no witness below $b\,p^{1/4}$. Then the root $\neg\Phi$ is a universal sentence true up to $b\,p^{1/4}$, so the axioms together with the root form a Normed$(b\,p^{1/4}, b)$ system. Now walk down the tree (Lemma 1):

```
Probe(p) {
  N := the root of p;
  while N is not a leaf {
    if N has two children {
      N := its left child if that is true, else its right child;
    } else {
      N := its only child;
      if N introduces a parameter u {
        give u the least value that keeps the branch positive;
      }
    }
  }
}
```

Here a node is *true* if its sentence holds under the values given so far, read with the horizons: universals up to $b\,p^{1/4}$, existentials below b, bounded sentences outright (Willard's condition II). The branch is *positive* if its nodes are true and each parameter has a value at most $b \cdot 2^{\Delta}$, where Δ counts the function symbols on the branch down to the node that introduced it (Eq. 19). Each function symbol at most doubles a value. So, by a nine-case induction, with one case for axioms and one for each of the eight elimination rules, the walk never gets stuck while every branch carries fewer than $\log_2(b\,p^{1/4}) - \log_2 b = \frac{1}{4}\log_2 p$ function symbols (Lemma 1). Willard's encoding spends more than five bits on each one, so a branch carries fewer than $\frac{1}{5}\log_2 p$ (Eq. 20).[4] The branch it finds cannot close, because a closing pair would be two bounded sentences that are both true (Lemma 2). So p has an open branch, and it is not a proof. ↯ Universal sentences are handled by permuting the quantifiers (2005, Appendix B). Theorem 3 follows, and Theorem 5 follows in the same way from Theorem 4.

Probe takes any purported refutation of the system and returns an open branch, with values for its parameters. Willard's consistency proof is therefore a machine that defeats each refutation by running it, in a universe barely larger than the refutation itself. That is self-justification ex machina

[3] The inequality needs $m \geq 3$. Every proof's Gödel number is far larger, since it is at least $32 \cdot 64^2$.

[4] Larger Gödel numbers only widen the margin. Willard notes that his theorems "would also be valid for uncompressed encoding methods" (2005, Appendix A). What would break this proof is a notation more compact than it allows, with four bits or fewer per function symbol along the deepest branch.

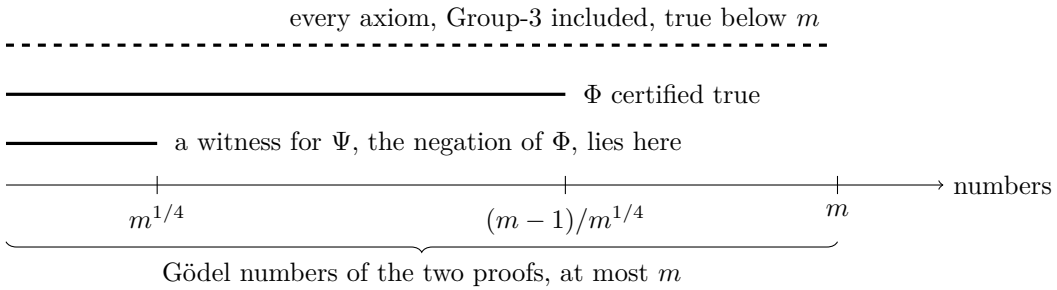


Figure 1: The two horizons in Willard's proof (schematic, not to scale). A universal theorem is certified out to its horizon, and an existential one finds its witness inside its horizon. When the horizons are short compared with the proofs, the two collide in the region where every axiom is true.

in the most literal sense. It is also the joke in the name C. The searcher hunts for a tree all of whose branches close, and the walk always finds one that stays open. $\mathrm{IS}_D(A)$ does not prove that the machine works. That part is external, and the system has only the axiom.

5 Transcripts Are Bigger Than Runs

Why should knowing be harder than watching? The natural proof of step (2) takes a proof s of $\ddot{\mathsf{G}}$ and produces the proof t of «$\mathsf{P}(\mathsf{P})$ halts» that exhibits s and checks it. Exhibiting s takes a term, and in Willard's system terms have a definite cost.

We measure cost in bits of proof. In Willard's language every function symbol at most doubles the largest of its arguments: $+$ and *Double* at most double it, and the other six never exceed it. In his encoding every function symbol occupies more than five bits. So each doubling costs more than five bits, and a term for a number v, which needs at least $\log_2 v$ doublings, costs more than $5 \log_2 v$ bits. The language and the encoding set this price. The deduction method makes it unavoidable, even for a number that a proof never names. Let the axioms be true universal sentences, as Willard's are over a base whose Π_1 theorems are true. Then a tableau proof p of an existential sentence writes down a term whose value is a witness, and that value is at most $p^{1/4}$ (Appendix A). Take «some y has $\mathrm{Log}(y) \geq n$». It names only n, but its witnesses are all at least 2^n, so each of its tableau proofs has $p \geq 2^{4n}$: at least four bits for each doubling. Only lemmas reused through cut could avoid the cost (Section 7). Nothing receives the bits. The price is a lower bound on the length of every proof that names the number or proves that it exists.

Lemma F (ours)**.** *In Willard's 2005 language and encoding, every tableau proof that contains a closed term of value $v \geq 2$ has a Gödel number greater than v^5.*

Proof. A closed term with L function symbols has value at most 2^L, because $+$ and *Double* at most double their largest argument and the other functions do not increase it. So a term of value v has at least $\log_2 v$ function symbols, and every branch through the node that contains it carries at least that many. Willard's encoding guarantees that every branch of a proof p carries fewer than $\frac{1}{5} \log_2 p$ function symbols (Eq. 20). Hence $\log_2 p > 5 \log_2 v$. □

So the transcript t is bigger than the fifth power of the run s it records. The natural proof of step (2) asks $\mathcal{F}$ to believe that $s \mapsto t$ is defined everywhere. Willard's systems cannot believe that of any function that outgrows every linear one. The reason lies in how a proof from universal axioms produces the numbers it claims exist.

Proposition G (Linear witnesses; ours)**.** *Let $\mathcal{F}$ be a set of true sentences of Willard's 2005 language, each bounded or Π_1, and let $\delta(\vec{x},\vec{y})$ be bounded. If $\mathcal{F}$ proves $\forall\vec{x}\,\exists\vec{y}\,\delta(\vec{x},\vec{y})$ by any sound deduction method, then there is a number k such that every $\vec{x}$ has a witness $\vec{y} \le k\,(x_1+\cdots+x_m+1)$.*

Appendix A has two proofs. One uses compactness and an initial segment closed under the function symbols. The other uses Herbrand's theorem, which presents the witnesses as terms in $\vec{x}$, built from the function symbols and from bounded searches. A term of depth d built from $+$, *Double* and the non-growth functions is at most 2^d times its largest argument. So k is like 2^d. That is exponential in the size of the proof, but it is fixed once the proof is, while $\vec{x}$ ranges over all numbers. With multiplication, a term of depth d can grow like x^{2^d}. That is the whole difference. In the units above, a proof that every input has an output buys a fixed number of doublings, the same for every input. A function that outgrows every linear one needs more doublings as its input grows, and no single proof has them.

Write $\mathrm{Num}(x,y)$ for a bounded formula saying that y is the Gödel number of the sentence $\bar{x}=\bar{x}$. For a bounded formula $\delta(x)$, let $\mathrm{Num}_\delta(x,y)$ say that y codes $\delta(\bar{x})$. Say that $\mathcal{F}$ *knows uniformly that it watches* δ if

$$\mathcal{F} \vdash \forall x\,[\delta(x) \supset \exists y\,\exists q\,(\mathrm{Num}_\delta(x,y) \wedge \mathrm{Prf}(y,q))].$$

This is Σ_1-completeness formalized with a free variable. Recursively axiomatized extensions of $I\Sigma_1$ prove it for their canonical provability predicate, and they derive the third derivability condition from it [Hájek–Pudlák 1993]. Willard's systems do not.

Corollary H (ours)**.** *Suppose every Π_1 theorem of A is true. Then $\mathrm{IS}_D(A)$ proves neither $\forall x\,\exists y\; y = x\cdot x$ nor $\forall x\,\exists y\,\mathrm{Num}(x,y)$. It cannot prove that every number has a written name, so it does not know uniformly that it watches even the formula $x=x$. The deduction method matters only through the truth of the axioms, so no sound method proves these sentences from them.*

Proof. Every axiom of $\mathrm{IS}_D(A)$ is true and is bounded or Π_1. For Groups 0–2 this follows from the hypothesis and Willard's remark that they "can be encoded as Π_1^* sentences" (2005, §5). For Group-3 it is Theorem D. So Proposition G applies, and only linear witnesses are available. But x^2 exceeds $k(x+1)$ once $x > 2k$. And the code of $\bar{x}=\bar{x}$ is at least $x^6/2$: the term $\bar{x}$ has at least $\log_2 x$ function symbols, one byte each, so the code is at least $32\cdot 64^{\log_2 x - 1}$ (Appendix A). Finally, knowing uniformly that it watches $x=x$ implies $\forall x\,\exists y\,\mathrm{Num}(x,y)$. □

Willard saw this in 1994 (TR 93-10):

> This means there will always be adequate memory for writing b_n's binary encoding when there is sufficient memory for storing a semantic tableaux proof of its existence (whereas the same is plainly not true *when multiplication is assumed to be a function*). For instance, multiplication allows one to establish the existence of a number a_{300} (in a proof with approximately 300 lines), whose binary encoding requires *more digits* than the number of atoms in the universe.

Here $b_{i+1} = b_i + b_i$ and $a_{i+1} = (a_i)^2$. A proof of n doublings names a number of about n bits, and a proof of n squarings names one of about 2^n bits. Later he put it as a comparison between a proof's Gödel number and the integer the proof shows to exist. Under additive naming the proof is always the bigger of the two, and under multiplicative naming it can be far smaller (2006a, Remark 3).

The faith step in Oberhoff's proof is a single closed sentence about the Gödel sentence, though, and there the totality claim is vacuous. If $\ddot{\mathsf{G}}$ is true, no proof of $\ddot{\mathsf{G}}$ exists, so "for every s there is a transcript" holds trivially. Proposition G cannot catch step (2) failing, so something else must.

6 Taking Off The Glasses

Oberhoff's §5 takes the glasses off $\ddot{\mathsf{G}}$ and finds "I am not provable" underneath. Let us take them off the faith step. One might think that Willard's system is self-justifying merely because nobody has managed to prove step (2) in it. In fact it cannot prove it, and the reason is Oberhoff's Theorem 6 run backwards.

Proposition I (ours)**.** *Let $\mathcal{F}$ have a sound and complete deduction method and a correct bounded proof predicate, and build $\ddot{\mathsf{G}}$ and H from them as in Appendix A. Suppose:*

(H1) $\mathcal{F}$ is Level(0−) consistent;

(H2) $\mathcal{F}$ watches;

(H3) $\mathcal{F}$ proves the true bounded sentences, and that $\leq$ is trichotomous at $\ulcorner\ddot{\mathsf{G}}\urcorner$;

(H4) $\mathcal{F}$ proves its own Level(1) consistency, for a bounded pairing of sentences with their negations that pairs $\ddot{\mathsf{G}}$ with H.

Then $\mathcal{F}$ proves that step (2) is equivalent to $\ddot{\mathsf{G}}$, and so is knowing that it watches at H. Yet $\mathcal{F}$ does not prove $\ddot{\mathsf{G}}$. So $\mathcal{F}$ proves neither step (2) nor $\mathsf{H} \supset \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner)$, although both are true. That $\mathcal{F}$ does not prove $\ddot{\mathsf{G}}$ uses only (H1), (H2), the soundness and completeness of the method, the correctness of the proof predicate, and the half of the fixed point that follows from one true bounded sentence ((FP⇒) of Fact N).

Proof. First, $\mathcal{F}$ proves step (3). By (H3) it proves that the pairing pairs $\ddot{\mathsf{G}}$ with H, and by (H4) two paired sentences never both have proofs. By (H3) and Fact N, $\mathcal{F}$ also proves the fixed point $\ddot{\mathsf{G}} \leftrightarrow \neg\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$. Now argue inside $\mathcal{F}$.

Suppose step (2):

⇒ If $\ddot{\mathsf{G}}$ had a proof, then so would H, by step (2), and step (3) forbids that.

⇒ So $\neg\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$, which by the fixed point is $\ddot{\mathsf{G}}$.

Suppose $\ddot{\mathsf{G}}$:

⇒ By the fixed point, $\ddot{\mathsf{G}}$ has no proof. So step (2), whose antecedent is $\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$, holds vacuously.

Both arguments use only theorems of $\mathcal{F}$ and logic, so Fact A turns them into a proof in $\mathcal{F}$ that step (2) is equivalent to $\ddot{\mathsf{G}}$. Lemma B(b) does the same for $\mathsf{H} \supset \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner)$. That $\mathcal{F}$ does not prove $\ddot{\mathsf{G}}$ is Oberhoff's §4, case 1.

Suppose $\mathcal{F}$ proves $\ddot{\mathsf{G}}$:

⇒ The proof predicate is correct, so $\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$ is a true Σ_1 sentence, and $\mathcal{F}$ proves it by watching. $\mathcal{F}$ sees $\mathsf{P}(\mathsf{P})$ halt.

⇒ By (FP⇒), $\mathcal{F}$ proves $\neg\ddot{\mathsf{G}}$ as well, and hence $0 = 1$ (Fact A), against (H1). ↯

Finally, $\ddot{\mathsf{G}}$ has no proof and the proof predicate is correct, so $\ddot{\mathsf{G}}$ is true. Step (2) is true because its antecedent is false, and $\mathsf{H} \supset \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner)$ is true because H is false. □

So the faith step is the Gödel sentence wearing glasses. The two are provably equivalent, and they are true and unprovable for the same reason. The step fails at the Σ_1 sentence that says $\mathsf{P}(\mathsf{P})$ halts, and at every sentence with the same proofs.[5] Willard's system meets every hypothesis.

[5] A Σ_1 rendering of «$\mathsf{P}(\mathsf{P})$ halts» with *different* proofs, such as $\exists x\,\exists h\,[\mathrm{SUBST}(\bar{n}, h) \wedge \mathrm{Prf}(h, x)]$ in the notation of Appendix A, is a different sentence, because provability is sensitive to codes. The proposition says nothing about it.

Corollary J (ours)**.** *Let A prove every Π_1 theorem of* PA+*, let every Π_1 theorem of A be true, and build $\ddot{\mathsf{G}}$ and H from Willard's own proof predicate for $\mathrm{IS}_D(A)$, with the conventions of Appendix A. Then, for semantic tableaux and for tableau lists, $\mathrm{IS}_D(A)$ proves neither $\ddot{\mathsf{G}}$, nor step (2), nor $\mathsf{H} \supset \mathrm{Prov}(\ulcorner \mathsf{H} \urcorner)$.*

Proof. (H1) follows from Theorem D, and (H2) is Lemma E. For (H3), Groups 0 and 1 prove the true bounded sentences. Trichotomy is a Π_1 theorem of PA+, so A proves it, and Group-2 hands it down. (H4) is the Group-3 axiom itself. The proof predicate is correct by Lemma E. Appendix A gives the details. □

The proposition is a contrapositive and claims no new method. Its sibling in Willard is 2001, Theorem 7.2: by the same fixed point, no consistent system of this kind proves canonical reflection for all its Π_1 sentences. In 1994 Willard had already set out the alternatives: "every axiom system either fails to prove its consistency or fails to meet the full Hilbert-Bernays conditions" (TR 93-10). The computational reading adds *which* condition fails, at *which* sentence, and that the failing instance is the Gödel sentence in disguise. What Willard adds, and no contrapositive can supply, is that systems on the far side of the alternative *exist*.

7 The Price Of Faith

If withholding faith is what lets $\ddot{\mathsf{W}}$ survive, the second theorem should return as soon as the system can afford the transcripts. Willard spent the second half of his career proving that it does.

One big number

Willard's 2005 paper ends with the experiment. Let 2_k^m be the least z whose k-fold base-2 logarithm is at least m, roughly a tower of k exponentials topped by m. Let $\Upsilon(k, m)$ be the true Σ_1 sentence "a number at least 2_k^m exists". $\mathrm{NS}_D^{k,m}(A)$ is $\mathrm{IS}_D(A)$ with $\Upsilon(k, m)$ added to Group-0, and with Group-3 adjusted to the larger system (Definitions 2 and 3). Both systems prove $\Upsilon(k, m)$, but only one has it for free.

Let $\mho$ be the bounded Gödel sentence "I have no short proof", that is, no proof p with $\mathrm{Log}^{k-1}(p) < m$ (Eq. 24). Lemma 4 is the faith step, bounded. For large k, Peano arithmetic in this language, PA+, proves a Π_1 sentence whose instance in Willard's proof is (27). It says that for every short proof p of $\mho$ and every n with $\mathrm{Log}^k(n) \geq m$ there is a transcript $q < n$ that checks p and proves "$\mho$ is provable". Lemma 4 rests on Lemma $3^\dagger$, a sketch showing that checking a proof is cheap. Because the transcript is bounded by n, (27) is Π_1, and the system inherits it through Group-2. What is missing is n, and $\Upsilon(k, m)$ supplies it.

Theorem K (Willard 2005, Theorem $6^\dagger$)**.** *For tableau lists there is a k_0 such that, for every $k > k_0$, $\mathrm{NS}_D^{k,m}$(PA+) is inconsistent.*

Willard's proof is a tableau list W of seven steps, and it is Oberhoff's Theorem 6 with the faith supplied:

- ⇒ (27), inherited from PA+: the faith step, below n.
- ⇒ The Group-3 axiom, at $\mho$ and its negation: step (3).
- ⇒ So no transcript exists: no p and q satisfy *CheckProof*$(\ulcorner \mho \urcorner, p, q)$. As printed, this step does not follow from the one before; see below.

$\Rightarrow$ With (27), no short proof of $\mho$ coexists with a number n that large. $\Upsilon(k,m)$, an axiom, supplies n, so no short proof of $\mho$ exists. That is $\mho$ itself.

$\Rightarrow$ Because $\Upsilon(k,m)$ cost nothing, W is short. For $k > k_0$ its Gödel number is below 2^m_{k-1}, so W is one of the short proofs that $\mho$ says do not exist. Watching W, the system proves that such a proof exists, and so it refutes $\mho$. $\lightning$

The dagger is ours. At our level of detail, the third arrow needs the transcript to prove the sentence that the consistency axiom pairs with $\mho$. As printed, it proves a different one. Appendix C sketches a repair, so we count the theorem as a sketch, although Willard prints a full proof. He also states, without proof, versions for semantic tableaux and other cut-free methods, for an empty base theory, and without the axioms that make addition and doubling total (Remark $8^\dagger$).

Why does the argument not run in $\mathrm{IS}_D(\mathrm{PA}+)$, which proves $\Upsilon(k,m)$ too? Willard's own engine answers; the observation is ours. Assume, as Theorem D does with $A = \mathrm{PA}+$, that every Π_1 theorem of PA+ is true. Then every axiom of $\mathrm{IS}_D(\mathrm{PA}+)$ is a true Π_1 sentence. Groups 0–2 "can be encoded as Π_1^* sentences" (2005, §5), and Group-3 is true by Theorem D. So the system is Normed$(a,1)$ for every $a > 1$. Let $p > 1$ be a proof of $\Upsilon(k,m)$, and take $a = p$, so that $p \leq \lceil a/1 \rceil + 1$. Then $\frac{1}{4}$-compactification puts a witness for $\Upsilon(k,m)$ at or below $p^{1/4}$. The least witness is 2^m_k, so $p \geq (2^m_k)^4$. In a list that uses $\Upsilon(k,m)$ as a lemma, the components up to the lemma form a list proof of it, and their bytes begin the whole list. So the whole list's Gödel number is at least $(2^m_k)^4$ too. That is far beyond the short proofs that $\mho$ speaks of. In Willard's words, "a Kleene-like Group-3 axiom has a different connotation when it refers to a short proof, referencing itself, than when it refers to a long such proof" (2005, §6). The asymmetry is exact. Adding a true Π_1 sentence instead of $\Upsilon(k,m)$ preserves consistency (2005, §$6^\dagger$). A universal axiom constrains. An existential axiom supplies a number, and the number is what the faith step was missing.

Squaring

Give a tableau system multiplication as a total function, and a proof of n lines can name 2^{2^n}, by $u_0 = 2$ and $u_{i+1} = u_i \cdot u_i$. Transcripts become affordable, and the second theorem returns. Willard located the mechanism himself (2000, §5):

> Many readers will smile with amusement when they learn the true reason that the Semantic Tableaux version of the Second Incompleteness Theorem breaks down when Multiplication is changed from a total function into a 3-way relation. It is essentially that Lemma 8 and its short proof then become no longer valid. ... Without it, our proof of the Semantic Tableaux version of the Second Incompleteness Theorem collapses entirely!

His Lemma 8 is the squaring construction: a short tableau proof that a number about as large as z exists. With it, the second theorem holds for semantic tableaux at Level$(0-)$. It holds for consistent finite extensions of $Q + V$, for an explicit Π_1 sentence V (2002a, Theorem 3.5; Theorem 3.4 shows that $Q + V$ is consistent). It also holds for every consistent extension of $I\Sigma_0$ in either of its two standard axiomatizations (2007, Theorem 6). That answers a question Paris and Wilkie asked in 1981. On the positive side, θ-compactification becomes "impossible to satisfy" once multiplication is a function symbol (2005, Remark $4^\dagger$).

Subroutines

A Hilbert-style proof can prove a lemma once and use it many times, so modus ponens is a subroutine call. A tableau must inline every use, and inlining can be astronomically expensive [Boolos 1984] (2002a, Theorem $2.2^\dagger$). With subroutines a little growth is amplified into a lot, as Willard saw early (TR 93-10):

These metaphors lose meaning when cuts are present because Addition can then be employed to *implicitly define* a sequence that grows actually faster than $a_0, a_1, a_2, a_3 \ldots$. Solovay's proof that IS(A) is incompatible with deductive cuts [So94] rests on this fast growth property.

He mapped the boundary by the complexity of the lemmas. Bounded lemmas are harmless (TR 93-10). Lemmas with one unbounded quantifier block are self-justifying (2005, Theorem 5). Lemmas with two blocks bring the second theorem back at Level(0−), for finite systems containing an explicit Π_1 theorem of Peano arithmetic (2004, Theorem $2^\dagger$). With unrestricted lemmas, that is, Hilbert deduction, the price falls to almost nothing. Solovay's theorem was communicated to Willard in 1994 and never published by Solovay (2006a, Theorem $2^\dagger$). Consider a consistent system that proves successor total and proves the Π_1 associative, commutative, distributive and identity laws for $+$ and $\times$ as relations. Solovay's theorem says that such a system cannot prove the absence of a Hilbert proof of $0 = 1$ from itself. The proof shortens definable cuts [Pudlák 1985]. A cut closed under successor yields, by short proofs that reuse lemmas, cuts closed under addition and then under multiplication, and on the last one the transcripts are affordable. So the Hilbert-style self-justifying systems give up successor itself. They name numbers by constants $C_1, C_2, \ldots$, and the naming convention is the system. Additive naming, $C_i = C_{i-1} + C_{i-1}$, works over a base with true Π_1 theorems and a bounded Hilbert proof predicate (2006a, Theorem 3). Multiplicative naming, $C_i = C_{i-1} \cdot C_{i-1}$, fails for every system that proves the Π_1 theorems of Peano arithmetic and meets Willard's concise-encoding condition (2006a, Theorem 4).

Two more prices complete the list. Excluded middle as a logical axiom at any node is a cut in disguise, and it makes the system inconsistent once its base proves the Π_1 ring laws for $+$ and $\times$ (2020, Theorem $4.5^\dagger$). And letting $\ddot{\mathsf{W}}$ assert more about itself, at Level(2^+), brings the second theorem back with addition alone, for finite systems containing an explicit Π_1 theorem of Peano arithmetic (2004, Theorem $1^\dagger$).

Proof system	Growth	Asserts	Outcome	Source
tableaux, or lists of Π_1/Σ_1 lemmas	$+$ total, $\times$ a relation	Level(1)	self-justifying	2005, Thms 3, 5
tableau lists, base PA+	the same, plus the axiom $\Upsilon(k, m)$, k large	Level(1)	inconsistent	2005, Thm $6^\dagger$
tableaux	$\times$ total	Level(0−)	second theorem	2002a, Thm 3.5; 2007, Thm 6
lists of Π_2/Σ_2 lemmas, finite systems	$+$ total	Level(0−)	second theorem	2004, Thm $2^\dagger$
tableaux, finite systems	$+$ total	Level(2^+)	second theorem	2004, Thm $1^\dagger$
tableaux with excluded middle	$+$ total, Π_1 ring laws for $+$ and $\times$	Level(1)	inconsistent	2020, Thm $4.5^\dagger$
Hilbert	successor total, ring laws for $+$, $\times$	Level(0−)	second theorem	Solovay†
Hilbert	additive names only	Level(0−)	self-justifying	2006a, Thm 3
Hilbert	multiplicative names	Level(0−)	second theorem	2006a, Thm 4

Willard's rule is to assert the weakest consistency in a negative result and the strongest in a positive one, so each row is as strong as its side allows. "Self-justifying" means that the system proves its own consistency and is consistent, given the hypothesis on its base theory.

8 Refactoring The Incompleteness Theorems, Again

Oberhoff's Theorem 7 is two theorems. Stated as properties of a system, they are these.

7a (Watching). *$\mathcal{F}$ proves every true Σ_1 sentence; in particular, the halting sentence of every halting computation.*

7b (Knowing that one watches). *$\mathcal{F} \vdash \sigma \supset \mathrm{Prov}(\ulcorner\sigma\urcorner)$ for every Σ_1 sentence σ; and, uniformly, $\mathcal{F}$ knows that it watches every bounded formula.*

7a is what we take from Oberhoff's Theorem 7. 7b, in both its forms, is our formulation of what his Theorem 6 takes on faith. 7a holds for Willard's 2005 and 2001 systems (Lemma E and Appendix B) and for every extension of Robinson's Q. 7b holds for recursively axiomatized extensions of $I\Sigma_1$ [Hájek–Pudlák 1993]. 7a gives the first incompleteness theorem, and Rosser's form also needs the loop fact, which Willard's systems inherit from their base. 7b is what the second theorem takes on faith, and it does not follow from 7a. Take Willard's 2005 system, with tableaux or tableau lists, over a base whose Π_1 theorems are true:

- the uniform form of 7b fails, since the system cannot prove that every number has a written name (Corollary H);
- when the base also proves every Π_1 theorem of PA+, 7b fails at the sentence H, «$\mathsf{P}(\mathsf{P})$ halts», and the failing instance is provably the Gödel sentence (Corollary J);
- yet 7b holds at every *true* Σ_1 sentence (Lemma B), so the failure is confined to a run that does not happen.

Willard's 2001 system behaves the same way, except at the closed sentence, where our argument leaves open which of two steps fails (Appendix B).

So Gödel's work splits into three parts, not two. The first is programming the integers (7a). The second is teaching the integers that they have been programmed (7b). The third is the victory laps. Willard's self-justifying systems live in the gap between the first part and the second, and the second theorem is a victory lap only for a runner who has already paid for 7b.

Self-reference is not where the trouble lies. The fixed point builds $\ddot{\mathsf{W}}$ in Willard's systems as easily as in Peano arithmetic. What costs something is *self-simulation*: running yourself in your head and writing down the result. The second incompleteness theorem needs both. The first needs only self-reference and watching, which is why Willard's systems obey it. Oberhoff found his classical and computational proofs "the exact same", and the same holds here. Willard's arithmetic (a least pair of contradictory proofs, values below $b \cdot 2^{\Delta}$, five bits per symbol) and the computational picture (a shortest refutation, the memory a proof occupies, a walk down a tree) are one argument. The arithmetic of Gödel numbers is the bookkeeping of memory. A refutation of $\ddot{\mathsf{W}}$ would have to establish the existence of a transcript-sized object inside a proof smaller than the object. The snake cannot swallow its tail, because its mouth is smaller than its body.

9 How To Build A Stable Genius

Here is the story in terms of halting. A consistent system that watches, with a correct proof predicate, has a blind spot. It does not prove «$\mathsf{P}(\mathsf{P})$ doesn't halt», because if it did, $\mathsf{P}(\mathsf{P})$ would halt, and the system, watching, would prove that too (the last part of Proposition I). The genius's axiom $\ddot{\mathsf{W}}$ is about a different run, that of $\mathsf{C}(\mathsf{C})$, the search for a contradiction in the genius itself. The bridge of Section 2, $\ddot{\mathsf{W}} \supset \ddot{\mathsf{G}}$, leads from the one run to the other: if $\mathsf{C}(\mathsf{C})$ doesn't halt, then $\mathsf{P}(\mathsf{P})$ doesn't halt. A system with both the axiom and the bridge would prove $\ddot{\mathsf{G}}$, inside its blind spot. So no system that meets (H1)–(H3) of Proposition I has both. The extensions of $I\Sigma_1$ in Proposition C prove all three steps of the bridge, because they can simulate themselves, and so the axiom makes

them inconsistent. Willard's system has the axiom. Of the bridge's three steps it proves the first and the third, and the faith step is the one it lacks (Corollary J).

Read as instructions, this is a recipe. To build a machine that knows it is consistent, keep what the first incompleteness theorem needs, and take away the bridge. Here are the ingredients, each with the result that requires it.

1. *Let it refer to itself.* This costs nothing. The fixed point needs one true bounded sentence and trichotomy at one number (Fact N).

2. *Let it watch.* Give it a finite set of true universal axioms that proves every true bounded sentence. Then it proves every true halting statement (Lemma E), and it reasons about algorithms in every sense the first incompleteness theorem needs. Over a base whose Π_1 theorems are true and include those of PA+, it obeys that theorem.

3. *Do not let it afford transcripts.* The natural proof of the faith step is a program, $s \mapsto t$, that turns a proof of $\ddot{\mathsf{G}}$ into the transcript that checks it, and t has more than five times the bits of s (Lemma F). So doublings must cost bits, in the units of Section 5: every number that a proof shows to exist must fit in the memory that the proof occupies, or, as Section 4 measures it, lie at or below the fourth root of the proof's Gödel number. Willard's machine meets this rule with an encoding that spends more than five bits on each function symbol, and with three prohibitions:

   - no function symbol that more than doubles its largest argument, so no multiplication;
   - no lemma with more than one unbounded quantifier block, and no excluded middle, which is a cut in disguise;
   - no axiom that hands it a large number for free.

   Section 7 shows that breaking any of the three, in the configurations treated there, brings back the second theorem or inconsistency.

4. *Then add the axiom*, at Level(1) and no higher. For finite systems containing an explicit Π_1 theorem of Peano arithmetic, asserting Level(2^+) brings the second theorem back with addition alone (2004, Theorem $1^\dagger$).

Followed in Willard's language, the recipe produces $\mathrm{IS}_D(A)$, which is consistent whenever the Π_1 theorems of A are true (Theorem D). Rule 3, with the truth of the base, is what makes rule 4 safe. Take a shortest refutation. Below the larger of its two Gödel numbers every axiom is true, the self-consistency axiom included, because the refutation is that axiom's least counterexample. And the refutation's two proofs cannot see far past their axioms. The existential one finds its witness close by, where the universal one has already certified everything. That is the collision of Section 4, and Probe is the program that finds it.

Rule 3 also fixes which programs the machine can certify. From a proof that every $\vec{x}$ has a $\vec{y}$ with $\delta(\vec{x}, \vec{y})$, Herbrand's theorem extracts finitely many terms, built from the function symbols and from bounded searches, such that at every input one of them is a witness (Appendix A). A program can try them in turn. So the function symbols are the instruction set of the machine's certified programs, and the programs are straight-line. Each makes a fixed number of calls, fixed by the proof and not by the input, and none has a loop whose length depends on the input. Bounded searches do loop, but they return nothing larger than their bounds. So with + and *Double* no certified program returns more than a fixed multiple of its input. That is Proposition G, read as a program.

A loop of input-dependent length would escape the bound: adding x to itself x times squares x. But to certify that such a loop halts on every input takes induction on an unbounded formula, and

the machine has no induction at all. It inherits the Π_1 theorems of its base, not the base's induction axioms. Induction on bounded formulas would not help either, by Parikh's theorem (Appendix A). Nor can a new predicate bring induction in. A true universal axiom, about any predicate, still holds in the initial segment built in Proposition G's first proof, where squaring fails (Appendix A). An axiom that proves squaring total must assert that numbers exist. It then hands the proof numbers that nothing written in the proof bounds, which rule 3 forbids. A transcript lies beyond the bound, and so does a name. The code of $\bar{x} = \bar{x}$ is at least $x^6/2$, so the machine cannot prove that every number can be written down (Corollary H). It writes down any number you hand it, by watching. It cannot prove that it could write them all. That is rule 3 in one line.

Moral of the story: you may declare yourself a very stable genius, and Willard's theorems will vouch for you, provided that your teachers told you only truths, you cannot multiply, and you reuse only lemmas with one unbounded quantifier block. Learn to multiply and, for the theories Willard treats, even proofs without lemmas are too much. Reuse lemmas with two blocks, and (by a sketch of Willard's) addition is already too much. Reuse any lemma you like, and (by Solovay's unpublished theorem) successor with the ring laws is too much. And if your teachers taught you Peano arithmetic and you reuse lemmas, you must never be told, as an axiom, that a big enough number exists (Theorem K, which we count as a sketch).

10 What The Genius Can Do

A program runs the same on a machine with the axiom as on one without it. What the axiom changes is what the machine will certify. Write $\mathrm{Cert}(\Phi)$ for the set of proofs that the checker of $\mathrm{IS}_D(A)$ accepts for Φ, that is, the q with $\mathit{SubstPrf}(\bar{n}_3, \ulcorner\Phi\urcorner, q)$. Group-3 gives programs one new law: if Φ is Π_1 and Pair pairs it with θ, then $\mathrm{Cert}(\Phi)$ and $\mathrm{Cert}(\theta)$ are not both inhabited. A *gate* is a program that guards an action. It takes a request, which is a claim together with a proof of it, checks the proof with the machine's own checker, and performs the action attached to the claim only if the check passes. Here is a gate whose safety rests on that law and on nothing else.

```
Gate(x, y, side, q) {
  z := x if side = 0, else y;
  if Pair(x, y) and q ∈ Cert(z) {
    emit (x, y, side);
  }
}
```

Read x as «transaction T never breaks the invariant», a Π_1 sentence, and y as its negation. The pair (x, y) is the key that names T. Side 0 commits T, and side 1 aborts it. Any number of copies of Gate may run, anywhere, with no shared state, on requests from any source: the machine's own searches, a rewritten successor's, an adversary's. A gate must honor every request that passes its test, since otherwise a gate that never emits anything would be safe for free.

Proposition L (ours)**.**

(a) $\mathrm{IS}_D(A)$ *proves that no two requests with the same key* (x, y) *and different sides both pass* Gate*'s test.*

(b) Let T *be a consistent, recursively axiomatized extension of* $I\Sigma_1$ *with Hilbert-style deduction, its canonical proof predicate, and a bounded pairing of* Π_1 *sentences with their negations. Let* Gate_T *test proofs in* T*. Then* T *does not prove the same of* Gate_T*.*

Proof. (a) Two such requests would be a counterexample to Group-3, which is an axiom. (b) The sentence is T's Level(1) consistency, and $I\Sigma_1$ proves that it implies T's Level(0−) consistency: by explosion, a proof of $0 = 1$ yields proofs of both sentences of a pair, as in the proof of Proposition C. By the second incompleteness theorem, T does not prove its Level(0−) consistency. □

So the stable genius can certify, from inside, a commit protocol that needs no coordination, and an ordinary system cannot certify its own copy of it at all. For a system that improves itself, this is the shape of the gain. A parent that approves its successors in advance, before it sees any of their proofs, knows that, as long as they keep its checker, no two certificates produced by any of them, or by itself, prove a Π_1 sentence and its negation. The searches can be replaced at will, by anything that produces certificates, and the proof of the guarantee does not grow with the number of replacements, because the guarantee concerns only the checker. Part (a) contains nothing beyond Group-3: the program is the axiom, read as a program. What is not trivial is that the machine survives the axiom (Theorem D).

What the genius does not gain is reflection. Everything it certifies is true, given the hypothesis on its base (Lemma E), but that is our theorem, not its own. Its own version would be the reflection principle at Φ, $\mathrm{Prov}(\ulcorner\Phi\urcorner) \supset \Phi$, which would let a program act on «Φ has a certificate» as on Φ. For a Π_1 sentence Φ the natural argument runs as follows. If Φ failed, a finite counterexample would exist. The machine would see it, so $\mathrm{dual}(\Phi)$ would have a certificate too, and the law above forbids two. The middle step is knowing that it watches at $\mathrm{dual}(\Phi)$, which is the faith step again.

Corollary M (ours)**.**

(a) For every Π_1 *sentence* Φ*,* $\mathrm{IS}_D(A)$ *proves* $[\mathrm{dual}(\Phi) \supset \mathrm{Prov}(\ulcorner\mathrm{dual}(\Phi)\urcorner)] \supset [\mathrm{Prov}(\ulcorner\Phi\urcorner) \supset \Phi]$*.*

(b) Under the hypotheses of Corollary J, $\mathrm{IS}_D(A)$ *does not prove* $\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner) \supset \ddot{\mathsf{G}}$*, although it is true.*

Proof. (a) Assume $\mathrm{dual}(\Phi) \supset \mathrm{Prov}(\ulcorner\mathrm{dual}(\Phi)\urcorner)$, $\mathrm{Prov}(\ulcorner\Phi\urcorner)$ and $\neg\Phi$. Then $\mathrm{dual}(\Phi)$ holds, being logically equivalent to $\neg\Phi$, and so does $\mathrm{Prov}(\ulcorner\mathrm{dual}(\Phi)\urcorner)$. Groups 0 and 1 prove the true bounded sentence $\mathrm{Pair}(\ulcorner\Phi\urcorner, \ulcorner\mathrm{dual}(\Phi)\urcorner)$, and Group-3 forbids the two proofs. So Φ. The argument is valid in every model of $\mathrm{IS}_D(A)$, and Fact A turns it into a proof. (b) By (H3) and Fact N, $\mathrm{IS}_D(A)$ proves (FP), which says that $\ddot{\mathsf{G}}$ is equivalent to $\neg\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$. So it proves $\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner) \supset \ddot{\mathsf{G}}$ equivalent to $\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner) \supset \neg\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$, hence to $\neg\mathrm{Prov}(\ulcorner\ddot{\mathsf{G}}\urcorner)$, hence to $\ddot{\mathsf{G}}$, which it does not prove (Corollary J). $\ddot{\mathsf{G}}$ is true, so the implication is true. □

So at the Gödel sentence, reflection is the product of self-consistency and faith, and the product is forbidden. In every consistent system that watches, with a correct proof predicate, and proves the fixed point, reflection at $\ddot{\mathsf{G}}$ is equivalent to $\ddot{\mathsf{G}}$, and so it is unprovable. Ordinary systems keep the faith and give up self-consistency. Willard's keep self-consistency and give up the faith. A stable genius proves that its certificates agree. It does not, in general, prove that they are true, although they are.

Put together, here is what the machine offers the programs that run on it, with the result behind each line.

Claim	Certified	Source
$q \in \mathrm{Cert}(\Phi)$, or $q \notin \mathrm{Cert}(\Phi)$, for a given q	yes	Groups 0 and 1; the proof predicate is bounded
$\mathrm{Cert}(\Phi)$ and $\mathrm{Cert}(\theta)$ are not both inhabited, when Pair pairs them	yes	Group-3; Theorem D
a given halting run halts	yes, run by run	Lemma E
$\bar{x} = \bar{x}$ has a proof, uniformly in x	no	Corollary H
every number has a square; every number has a written name	neither	Corollary H
had $\mathsf{P}(\mathsf{P})$ halted, a proof of that would exist	no	Corollary J
a proof of $\ddot{\mathsf{G}}$ makes $\ddot{\mathsf{G}}$ true	no	Corollary M

Willard enforces rule 3 by limiting growth. It can also be enforced by accounting. Let each node of a certificate cost one unit of a resource that programs can neither copy nor create, so that a program can hold no certificate larger than its budget. A refutation on a budget of n that uses the axiom must hand it a certificate of a refutation. That certificate has at most n nodes and must name every unit that the refutation it describes spends, so it describes a refutation on a smaller budget. A least refutation therefore cannot use the axiom, which is safe if the rest of the system is consistent. The calculus $\lambda_0^{\mathrm{cert}}$ in the `sjas` repository is built on this plan [sjas 2026]. There the restriction falls on certificates alone, ordinary data keep the strength of arithmetic in all finite types, and the axiom speaks of the certificates that a program holds.

What it does not buy. Willard was more careful about the limits of his results than most of his readers have been.

- *The consistency is relative.* The systems are exactly as trustworthy as their base. In Willard's words, if A is Peano arithmetic, "its consistency will be as certain as that of Peano Arithmetic!" (2001, §2).
- *The proof is one line.* A system that proves its consistency from its own axiom $\ddot{\mathsf{W}}$ has "essentially a 1-line proof" (2011, Remark 6.16). That is faith, not justification. The justification is Probe, which lives outside, in the metatheory. Willard put the objection against himself: "Is it not almost cheating . . . ?" (2016, §8). His answer, Platonic Stability, rests on a conjecture (2016, Conjecture $6.6^{\dagger}$).
- *The words are ours.* Watching and knowing, the conventions of Appendix A, Lemmas B, E and F, Propositions G, I and L and their corollaries, the program C as a reading of Willard's Group-3, the horizon picture, the bridge and the recipe of Section 9, the program Gate, and the repair in Appendix C are ours. Willard's own words for the mechanism are "adequate memory", and the comparison between a proof's Gödel number and the integer it proves to exist.

(The cake promised in Oberhoff's abstract is still on its way. We cannot prove that it exists, because it is one multiplication away.)

A The Fine Print

A.1 Languages, consistency and arithmetization

The 2005 language. Willard's *U-Grounding* language has constants $\bar{c}_0$ and $\bar{c}_1$ for 0 and 1, the growth functions $+$ and *Double*, and six *grounding* functions with $F(\vec{a}) \leq \max(\vec{a})$. These are

truncated subtraction, integer division, *Root*, *Maximum*, *Logarithm* and *Count* (2005, §2). Each $n \geq 2$ is named by its *U-Grounded Binary Representation* $\bar{n}$. This is a term with at most $2 \log_2 n$ occurrences of + and *Double*, applied to 1. Bounded quantifiers have the forms $\forall v \leq t$ and $\exists v \leq t$. Willard's classes Δ_0^*, Π_1^* and Σ_1^* are written "bounded", Π_1 and Σ_1 in the main text. A negation in front of unbounded quantifiers yields a sentence in neither class, although it is logically equivalent to one (2005, Remark 5). The graph of multiplication is a Δ_0^* formula (Eq. 4). Multiplication by any fixed constant is total, through iterated addition. What is withheld is the uniform operation $x \cdot y$. PA+ is Peano arithmetic with the U-Grounding function symbols (2005, §6). The 2001 language is in Appendix B.

Consistency. For an axiom system α with deduction method D, Level(0−) consistency means that there is no D-proof of $0 = 1$ from α. Level(1) consistency means that no Π_1^* sentence and its negation both have D-proofs from α (2005, §2). The negation is made precise by Pair below. The levels agree in strong systems and come apart in weak ones. Willard asserts the weakest level in his negative results and the strongest in his positive ones (2002c).

Arithmetization. In each configuration here, $\mathrm{Prf}_\alpha^D(x, y)$, "$y$ codes a D-proof from α of the sentence coded x", is a bounded formula, and $\mathrm{Prov}_\alpha^D(x) := \exists y \, \mathrm{Prf}_\alpha^D(x, y)$. The other syntactic predicates used here also have bounded definitions: substitution, Num, the prenex rewriting and Pair. This is Willard's arithmetization of syntax. It runs through Wrathall's characterization of the linear-time hierarchy (2001, Theorems C.1–C.2† and C.3; 2005, §3). It is Oberhoff's Theorem 7 transposed to languages without total multiplication, and, like him, we take it as given. For the proof-searching programs of this essay, «A(A) halts» is formalized as "the search succeeds". That is an existential statement about Prf.

A.2 Encoding, prenex form and negation

Encoding. Proofs are encoded as in 2005, Appendix A. A proof is a string of six-bit bytes read as an integer in base 64. Each of the 24 symbols has a code between 32 and 55, so its leading bit is 1. A tableau list is the concatenation of its components. A sentence is numbered by the value of its own byte string, as in 2001, Appendix B, with the first byte in the highest place. Two consequences are used throughout. First, a string of N bytes that begins with a symbol byte has value at least $32 \cdot 64^{N-1}$, and every proof and every sentence begins with one. Second, a sentence that occurs in a proof has a Gödel number no larger than the proof's, because a contiguous substring of a base-64 numeral has no larger value. Every axiom used in a tableau occurs in it as a node (2005, §2).

Prenex form. The root of a tableau proof of Φ "stores the sentence $\neg\Phi$ (rewritten in Prenex* normal form)" (2005, §2). Willard does not fix the rewriting, so we fix it. Let $\Theta = Q_1 v_1 \ldots Q_k v_k \, \delta$ be a prenex sentence with δ bounded. Let $\mathrm{dual}(\Theta) := \bar{Q}_1 v_1 \ldots \bar{Q}_k v_k \, \neg\delta$, where each quantifier is exchanged for the other. Let $\mathrm{pnx}(\Theta) := \Theta$ and $\mathrm{pnx}(\neg\Theta) := \mathrm{dual}(\mathrm{pnx}(\Theta))$. Willard's arguments use only that the root is a prenex sentence equivalent to $\neg\Phi$, in the class of that sentence, so they hold under this convention. He also states that they hold without the prenex requirement (2005, §2†). For a prenex Φ,

$$\mathrm{pnx}(\neg\neg\Phi) = \mathrm{dual}(\mathrm{dual}(\Phi)) = \mathrm{pnx}(\neg \, \mathrm{dual}(\Phi)).$$

So the syntactic negation $\neg\Phi$ and the prenex sentence $\mathrm{dual}(\Phi)$ have exactly the same tableau proofs. They also have the same tableau-list proofs, since a list proves the conclusion of its last component.

Negation in the consistency axiom. Level(1) consistency is stated with a bounded formula $\mathrm{Pair}(x, y)$, "x is the Gödel number of a Π_1^* sentence and y is x's negation" (2005, §3):

$$\forall x\, \forall y\, \forall p\, \forall q\, \neg[\mathrm{Pair}(x, y) \wedge \mathrm{Prf}(x, p) \wedge \mathrm{Prf}(y, q)].$$

Willard's text uses two candidates for the negation. One is the string $\neg f$ (his Eq. 28). The other is "the Σ_1^* theorem that is Φ's negation" in the proof of Theorem 1, that is, $\mathrm{dual}(\Phi)$. We let $\mathrm{Pair}(x, y)$ hold when x codes a Π_1^* sentence Φ and y codes a sentence θ with $\mathrm{pnx}(\neg\theta) = \mathrm{pnx}(\neg\neg\Phi)$. Both candidates qualify, and every sentence Pair accepts has exactly the tableau proofs of $\mathrm{dual}(\Phi)$. The first arrow of Theorem 1's proof in §4 needs this, because witnesses are defined only for Σ_1^* sentences. With the rewriting left open, Willard's "such a triple (j_1, j_2, m) will clearly exist" would pass over the step.

A.3 The Gödel sentence and the fixed point

Let $\mathrm{SUBST}_0(g, h)$ be a bounded formula defining the graph of substitution: h codes the result of substituting the numeral of g into the formula coded g. We use it in least-witness form,

$$\mathrm{SUBST}(g, h) := \mathrm{SUBST}_0(g, h) \wedge \forall h' \leq h\, [h' = h \vee \neg\mathrm{SUBST}_0(g, h')].$$

This formula is bounded, and in the standard model it holds exactly when SUBST_0 does. Let n be the Gödel number of

$$\Gamma(g) := \forall x\, \forall h\, [\mathrm{SUBST}(g, h) \supset \neg\mathrm{Prf}(h, x)],$$

and let $\ddot{\mathsf{G}} := \Gamma(\bar{n})$ and $c := \ulcorner\ddot{\mathsf{G}}\urcorner$. So $\ddot{\mathsf{G}}$ is Π_1, and $\mathrm{SUBST}(\bar{n}, \bar{c})$ is a true bounded sentence. The *halting sentence* is the Σ_1 sentence

$$\mathsf{H} := \mathrm{dual}(\ddot{\mathsf{G}}) = \exists x\, \exists h\, \neg[\mathrm{SUBST}(\bar{n}, h) \supset \neg\mathrm{Prf}(h, x)].$$

It says that some x proves the sentence that substitution produces. With $c' := \ulcorner\mathsf{H}\urcorner$, $\mathrm{Pair}(\bar{c}, \bar{c}\,')$ is a true bounded sentence, and H has exactly the proofs of $\neg\ddot{\mathsf{G}}$.

Our $\ddot{\mathsf{G}}$ differs from Willard's fixed point (2001, Eq. 63) in two ways. It has no bound $h \leq x$, just as his own bounded Gödel sentence $\mho$ has none (2005, Eq. 24). And its substitution formula is in least-witness form. Together these make the fixed point provable from trichotomy alone.

Fact N (Fixed point; after Willard 2001, Theorem 7.2)**.** *Let D be sound and complete for α, let* Prf *be bounded, and let α prove* $\mathrm{SUBST}(\bar{n}, \bar{c})$*. Then*

$$\alpha \vdash \ddot{\mathsf{G}} \supset \forall y\, \neg\mathrm{Prf}(\bar{c}, y). \tag{FP$\Rightarrow$}$$

If α also proves $\forall h\, (h \leq \bar{c} \vee \bar{c} \leq h)$, then it proves (U) $\forall h\, [\mathrm{SUBST}(\bar{n}, h) \supset h = \bar{c}]$ *and*

$$\alpha \vdash \ddot{\mathsf{G}} \leftrightarrow \forall y\, \neg\mathrm{Prf}(\bar{c}, y). \tag{FP}$$

Proof. (FP$\Rightarrow$): instantiate $\ddot{\mathsf{G}}$ at $x = y$ and $h = \bar{c}$. With $\mathrm{SUBST}(\bar{n}, \bar{c})$ this gives $\neg\mathrm{Prf}(\bar{c}, y)$.

(U): suppose $\mathrm{SUBST}(\bar{n}, h)$. If $h \leq \bar{c}$, the second conjunct of $\mathrm{SUBST}(\bar{n}, \bar{c})$ at $h' = h$ gives $h = \bar{c}$ or $\neg\mathrm{SUBST}_0(\bar{n}, h)$, and the first conjunct of $\mathrm{SUBST}(\bar{n}, h)$ excludes the latter. If $\bar{c} \leq h$, the second conjunct of $\mathrm{SUBST}(\bar{n}, h)$ at $h' = \bar{c}$ gives $\bar{c} = h$ or $\neg\mathrm{SUBST}_0(\bar{n}, \bar{c})$, and the first conjunct of $\mathrm{SUBST}(\bar{n}, \bar{c})$ excludes the latter.

(FP$\Leftarrow$): given x and h with $\mathrm{SUBST}(\bar{n}, h)$, (U) gives $h = \bar{c}$, and the hypothesis gives $\neg\mathrm{Prf}(\bar{c}, x)$.

Each implication holds in every model of α, given theorems of α. Applying Fact A repeatedly gives its proof. □

H is logically equivalent to $\neg\ddot{\mathsf{G}}$, so α proves them equivalent. That does not give $\alpha \vdash \mathrm{Prov}(\ulcorner\neg\ddot{\mathsf{G}}\urcorner) \leftrightarrow \mathrm{Prov}(\ulcorner\mathsf{H}\urcorner)$, which would be a derivability condition, the very thing in question. What the conventions give instead is that $\neg\ddot{\mathsf{G}}$ and H have the same proofs, and that Pair pairs $\ddot{\mathsf{G}}$ with both.

A.4 Willard's 2005 system: proofs for Sections 4 to 6

The proof predicate. Let UNION(A) be Groups 0–2. $\mathit{ExPrf}^{D}_{\mathrm{UNION}(A)}(h,t,p)$ says that p is a D-proof of t from UNION(A) plus the sentence coded h. Willard's Eq. (7) is

$$\mathit{SubstPrf}(g,t,p) := \mathrm{Prf}^{D}_{\mathrm{UNION}(A)}(t,p) \vee \exists h \le p\,[\mathrm{SUBST}(g,h) \wedge \mathit{ExPrf}^{D}_{\mathrm{UNION}(A)}(h,t,p)].$$

Group-3, his Eq. (8), is $\Gamma_3(\bar{n}_3)$, where n_3 is the Gödel number of

$$\Gamma_3(g) := \forall x\,\forall y\,\forall p\,\forall q\,\neg[\mathrm{Pair}(x,y) \wedge \mathit{SubstPrf}(g,x,p) \wedge \mathit{SubstPrf}(g,y,q)].$$

This is «$\mathsf{C}(\mathsf{C})$ doesn't halt», with g standing for C. The proof predicate of $\mathrm{IS}_D(A)$ is $\mathit{SubstPrf}(\bar{n}_3,\cdot,\cdot)$, which is bounded (2005, §3).

Lemma E: correctness. By the arithmetization, SUBST, $\mathrm{Prf}^{D}_{\mathrm{UNION}(A)}$ and *ExPrf* are correct. A D-proof p from $\mathrm{IS}_D(A)$ that does not use Group-3 satisfies the first disjunct of Eq. (7). One that does use it contains Group-3 as a node. So by the encoding, Group-3's code h_3 is at most p, and the second disjunct holds with $h = h_3$. Conversely, each disjunct describes a D-proof from $\mathrm{IS}_D(A)$, since in the standard model $\mathrm{SUBST}(\bar{n}_3,h)$ holds only for $h = h_3$.

Lemma E: truth. Groups 0 and 1 are true. In Willard's words, "it is obvious that all $\mathrm{IS}_D(A)$'s Group-0 and 1 axioms are valid under the standard model" (2005, §5). Group-2 is true by hypothesis, and Group-3 is true by Theorem D. That theorem's proof rules out D-proofs of Φ and dual(Φ), and by the conventions above, every sentence that Pair accepts has the proofs of dual(Φ). A sound method proves only truths from true axioms.

Fact O (Inheritance)**.** *If A proves a Π_1 sentence Φ, then so does* $\mathrm{IS}_D(A)$.

Proof. Let p be the proof. Groups 0 and 1 prove the true bounded sentence $\mathrm{Prf}_A(\ulcorner\Phi\urcorner,\bar{p})$, and with Φ's Group-2 axiom, Fact A gives Φ. □

The first incompleteness theorem. Let A prove every Π_1 theorem of PA+, and let every Π_1 theorem of A be true. Then $\ddot{\mathsf{G}}$ is unprovable by Corollary J. $\neg\ddot{\mathsf{G}}$ and H are false, so they are unprovable, because $\mathrm{IS}_D(A)$ is honest. Rosser's version goes through as in Oberhoff. The finite searches he writes out are true bounded facts, which Groups 0 and 1 prove. In case 2, $\mathsf{B}(\mathsf{B})$ enters its loop at a definite step, so PA+ proves the Π_1 sentence «$\mathsf{B}(\mathsf{B})$ doesn't halt», and Fact O hands it to $\mathrm{IS}_D(A)$.

Corollary J: the hypotheses. The proof predicate is bounded and correct. (H1): a proof of $0 = 1$ would yield proofs of every sentence, by Fact A, and in particular of a Π_1 sentence and its negation. Group-3 is true, so no such pair exists. (H2) is Lemma E. (H3): Groups 0 and 1 prove the true bounded sentences, among them $\mathrm{SUBST}(\bar{n},\bar{c})$ and $\mathrm{Pair}(\bar{c},\bar{c}\,')$. Trichotomy is a Π_1 theorem of Peano arithmetic and so of PA+. So A proves it, by hypothesis, and Fact O hands it down. (H4) is literally Eq. (8). $\mathrm{IS}_D(A)$ is never asked to prove that its Group-3 formula is its own proof predicate. That correctness is used only in the metatheory.

Witnesses in tableau proofs. Let the axioms be true universal sentences, and let p be a tableau proof of $\exists y\, \varphi(y)$, with φ bounded. Walk down p as Probe does: at each fork step to a true child, and give each new parameter the least value that makes its sentence true. Axioms are true, and each rule turns true sentences into true ones. The root, which says that no witness exists, is false, and its instance $\neg\varphi(t)$ is false exactly when the value of t is a witness. The walk must meet such an instance, because its branch closes with a bounded sentence and its negation (2005, Lemma 2), and these cannot both be true. Because the axioms and the root are universal, every parameter is introduced by a bounded quantifier, «some $z \le s$ has ψ», and its value is at most that of s, which is written higher on the branch. Each function symbol at most doubles. So the value of t is at most 2^L, where L counts the function symbols on the branch (2005, Eq. 19), and $2^L < p^{1/5}$ (Eq. 20). For a single tableau, then, a number that the proof never names costs as much as one it names (Lemma F). Willard's Theorems 2 and 4 give the bound $b\,p^{1/4}$ for tableaux and for tableau lists, with axioms true only up to a horizon and existential axioms witnessed below b. Section 7 applies them with $b = 1$.

Proposition G: proof. Write $|\vec{x}|$ for $x_1 + \cdots + x_m + 1$, and $\bar{k} \cdot t$ for the term $t + \cdots + t$ with k summands. Suppose that no k works. Then for each k some $\vec{x}$ has no witness $\vec{y} \le k|\vec{x}|$. Add new constants $\vec{c}$, and consider every finite subset of

$$\mathrm{Th}(\mathbb{N}) \cup \{\forall \vec{y} \le \bar{k} \cdot |\vec{c}|\ \neg\delta(\vec{c}, \vec{y}) : k \in \mathbb{N}\}.$$

$\mathbb{N}$ satisfies it, if $\vec{c}$ is interpreted as such an $\vec{x}$ for the largest k involved. By compactness there is a model M of $\mathrm{Th}(\mathbb{N})$ with elements $\vec{a}$ such that $M \models \forall \vec{y} \le \bar{k} \cdot |\vec{a}|\ \neg\delta(\vec{a}, \vec{y})$ for every $k \in \mathbb{N}$. Let

$$I := \{\, z \in M : z \le \bar{k} \cdot |\vec{a}| \text{ for some } k \in \mathbb{N} \,\}.$$

The following facts hold in M, because the sentences that express them are true in $\mathbb{N}$.

- $\le$ is transitive, so I is an initial segment.
- $x_i \le x_1 + \cdots + x_m + \bar{c}_1$, so each a_i is in I.
- $\bar{n} \le \bar{n} \cdot u$ for $u \ge 1$, so I contains every $\bar{n}$.
- $\bar{i} \cdot u \le \bar{j} \cdot u$ for $i \le j$, so any finitely many elements of I lie under one bound $\bar{k} \cdot |\vec{a}|$.
- If $x \le \bar{i} \cdot u$ and $y \le \bar{j} \cdot u$, then $x + y \le \overline{i+j} \cdot u$ and $\mathit{Double}(x) \le \overline{2i} \cdot u$. So I is closed under $+$ and *Double*.

The grounding functions satisfy $F(\vec{z}) \le \max(\vec{z})$, and their arguments lie under a common bound, so I is closed under them too. Hence I is a substructure of M and an initial segment of it. A bounded formula therefore has the same truth value in I as in M, at arguments from I. Its terms take the same values, and its quantifiers range over initial segments of I. So a bounded sentence of $\mathcal{F}$ is true in M, and hence in I. A Π_1 sentence $\forall \vec{z}\, \beta(\vec{z})$ of $\mathcal{F}$ is true in M, so β holds at every tuple from I, in M and hence in I. So $I \models \mathcal{F}$, hence $I \models \forall \vec{x}\, \exists \vec{y}\, \delta$. Then some $\vec{y} \in I$ satisfies $\delta(\vec{a}, \vec{y})$ in I, and hence in M. But $\vec{y} \le \bar{k} \cdot |\vec{a}|$ for some $k \in \mathbb{N}$, which contradicts the choice of $\vec{a}$. ↯

Induction on bounded formulas. The proposition stays true if $\mathcal{F}$ also contains the induction axioms for bounded formulas, because I satisfies them. Suppose a bounded formula, with parameters in I, holds at 0, passes from each element of I to its successor, and fails at some $z \in I$. Its least failure in M exists, because $M \models \mathrm{Th}(\mathbb{N})$. That failure is at most z, so it lies in I. It is not 0, so

it is the successor of an element of I at which the formula holds, and the formula passes to the successor. ↯ This is Parikh's theorem [Parikh 1971]. The proof also survives new predicate symbols. Take $\mathrm{Th}(\mathbb{N})$ in the larger language, and true universal axioms about the new predicates hold in I as before. Induction on Σ_1 formulas is another matter. Added to $\mathrm{IS}_D(A)$, it proves $\forall x\,\exists y\; y = x \cdot x$ by induction on x, because $(x+1)\cdot(x+1) = x\cdot x + \mathit{Double}(x) + 1$.

Proposition G: a second proof, by Herbrand's theorem. This proof computes the witnesses. Write each bounded quantifier as $\exists v \le t\,\varphi(v,\vec{z})$, a universal one as $\neg\exists v \le t\,\neg\varphi$. Working from the innermost quantifier of $\mathcal{F}$ and δ outwards, add for each one a function symbol $\mu(\vec{z})$, read as the least $v \le t(\vec{z})$ with φ, or 0 if there is none, together with the true universal axioms $\mu(\vec{z}) \le t(\vec{z})$ and $v \le t(\vec{z}) \wedge \varphi(v,\vec{z}) \supset \varphi(\mu(\vec{z}),\vec{z})$. They make $\exists v \le t\,\varphi$ provably equivalent to the quantifier-free $\mu \le t \wedge \varphi(\mu)$. So $\mathcal{F}$ becomes a universal theory U, true in $\mathbb{N}$, which proves $\forall\vec{x}\,\exists\vec{y}\,\delta^*$ for a quantifier-free δ^* equivalent to δ. By Herbrand's theorem for universal theories [Buss 1998], there are tuples of terms $\vec{t}_1,\ldots,\vec{t}_n$ with

$$U \vdash \forall\vec{x}\,[\delta^*(\vec{x},\vec{t}_1(\vec{x})) \vee \cdots \vee \delta^*(\vec{x},\vec{t}_n(\vec{x}))].$$

U is true in $\mathbb{N}$, so at every input one of the $\vec{t}_i(\vec{x})$ is a witness. The symbols $+$ and *Double* at most double their largest argument, the grounding functions do not exceed it, and each μ is bounded by a term of the original language in its arguments. So, by induction on terms, each term s has a d with $s(\vec{x}) \le 2^d \max(\vec{x},1)$ for all $\vec{x}$. The program that evaluates $\vec{t}_1(\vec{x}),\ldots,\vec{t}_n(\vec{x})$ in turn and returns the first that satisfies δ finds a witness within the bound of the proposition, with $k = 2^d$ for the largest of these d.

B The 2001 System

The system. Willard's 2001 system $\mathrm{IS}(A)$ is written in a different language. It has a constant $\bar{n}$ for every number n, and seven grounding functions: truncated subtraction, integer division, predecessor, maximum, logarithm, root and *Count*. Its classes are written Δ_0^-, Π_1^- and Σ_1^- (2001, §2). Addition is not a function symbol. Group-1 asserts that addition is total by the Π_2^- axiom $\forall x\,\forall y\,\exists z\; z \dot{-} x = y$ (Eq. 6). The deduction method is semantic tableaux. Group-3 asserts Level(0−) consistency. It is $\Theta(\bar{N})$, with

$$\Theta(g) := \forall y\,\neg \mathit{SubstSemPrf}_{\mathrm{UNION}(A)}(\ulcorner 0 = 1 \urcorner, y, g).$$

Here $\mathit{SubstSemPrf}_{\mathrm{UNION}(A)}(s,y,g)$ says that y is a tableau proof of s from $\mathrm{UNION}(A)$ together with the sentence h such that $\mathrm{SUBST}(g,h)$ (2001, Appendix B, Eqs. 86–87). Its Δ_0^- definition is Theorem C.3. That theorem obtains it from analogs of Theorem C.1, which are called routine and are not written out, and from Theorem C.2†. Willard assumes throughout that A extends Peano arithmetic. A is *regularly consistent* if Prf_A has a Δ_0^- encoding and every Group-2 axiom is true (2001, §2).

Theorem P (Willard 2001, Theorem 4.3)**.** *If A is regularly consistent, then* $\mathrm{IS}(A)$ *is Level(0−) consistent.*

Willard's proof is an eleven-case induction by contradiction. He later wrote that such proofs are "quite difficult to visualize", that this one "had this problem", and he replaced it with the engine of §4 (2005, Remark 2).

What carries over. Let $A \supseteq$ PAX be regularly consistent, where PAX is Peano arithmetic in the 2001 language. Let $\ddot{\mathsf{G}}$ and H be built from the proof predicate of IS(A)'s Group-3 axiom, under the conventions of Appendix A.

- *Watching.* Since $A \supseteq$ PAX, Group-2 (2001, Eq. 7) hands IS(A) every Π_1^- theorem of PAX. Willard notes that a system that can prove all of these "obviously has a capacity to prove every Δ_0^- sentence which is true" (2001, §7). IS(A) also names every number by a constant. So it proves the true Δ_0^- instance of a true Σ_1^- sentence at its witnesses, and Fact A adds the quantifiers.

- *The fixed point.* Among the true Δ_0^- sentences that IS(A) proves is SUBST($\bar{n}, \bar{c}$). Trichotomy at $\bar{c}$ is a Π_1^- theorem of PAX, which Group-2 hands down. So Fact N gives (FP) in IS(A), and (H3) of Proposition I holds.

- *Correctness.* Write the Δ_0^- definition in the form of 2005's Eq. (7):

$$SemPrf_{\mathrm{UNION}(A)}(s, y) \vee \exists h \leq y\, [\mathrm{SUBST}(g, h) \wedge SemPrf_{\mathrm{UNION}(A)+h}(s, y)].$$

  A proof y that uses no Group-3 axiom satisfies the first disjunct. A proof that uses Group-3 has $\Theta(\bar{N})$ as a node: the nodes other than the root "are either axioms of α or deductions from higher nodes" (2001, §4). Each node is written as a curly bracket followed by its sentence (2001, Appendix B), so the axiom's byte string is a contiguous part of y. So its code h_N is at most y, and the second disjunct holds with $h = h_N$. Conversely, the first disjunct describes a proof from UNION(A) $\subseteq$ IS(A). The second describes a proof from UNION(A) plus the sentence h with SUBST($\bar{N}, h$), and in the standard model that sentence is only h_N.

- *Honesty.* Group-0 consists of true atomic facts, Group-1 is true, and Group-2 is true by regular consistency. Group-3 says, by correctness, that no tableau proof of $0 = 1$ exists, and this is true by Theorem P.

- *The first incompleteness theorem.* The argument that $\mathcal{F}$ does not prove $\ddot{\mathsf{G}}$, in Proposition I, needs only Level(0−) consistency (Theorem P), watching, correctness, the soundness and completeness of tableaux, and (FP⇒). So IS(A) does not prove $\ddot{\mathsf{G}}$. Honesty excludes $\neg\ddot{\mathsf{G}}$. Rosser's loop sentence is a Π_1^- theorem of PAX, handed down by the Group-2 axiom of 2001 (Eq. 7). This is what TR 93-10 means by "Godel's First but not Second".

- *Linear witnesses.* Proposition G holds for true bounded and Π_1^- sentences together with Eq. (6). Eq. (6) says only that $z \dot{-} x = y$ has a solution, but its sharpening $z \dot{-} x = y \wedge x \leq z$ is bounded. The true sentences saying that the sharpening is total and single-valued hold in every model of Th($\mathbb{N}$). So expand $\mathbb{N}$ and M by the function it defines, and write the bounds $\bar{k} \cdot |\vec{x}|$ with it. The argument then runs verbatim. I is closed under the defined addition and under the non-growth functions, and contains every constant. Eq. (6) holds in I, because $z = x + y$ lies in I when x and y do.

- *The uniform faith step fails.* All axioms are true, and each is bounded, Π_1^- (Group-3 by Theorem C.3), or Eq. (6). The constant $\bar{x}$ takes $\lceil \log_{32}(x+1) \rceil + 1$ bytes (2001, Appendix B), so the code of $\bar{x} = \bar{x}$ is at least $32\,(x+1)^{6/5}$. That eventually exceeds $k(x+1)$, as in Corollary H.

What does not carry over. Group-3 asserts only Level(0−) consistency, so step (3) is no longer free.

Corollary Q (ours)**.** IS(A) *does not prove both step (2) and the Level(0−) form of step (3),*

$$\forall p\,\forall q\,[\mathrm{Prf}(\bar{c},p)\wedge\mathrm{Prf}(\bar{c}\,',q)\supset\exists r\,\mathrm{Prf}(\ulcorner 0=1\urcorner,r)].$$

Proof. With Group-3, $\forall r\,\neg\mathrm{Prf}(\ulcorner 0=1\urcorner,r)$, these two sentences would give step (3), $\neg[\mathrm{Prov}(\bar{c})\wedge\mathrm{Prov}(\bar{c}\,')]$. That is all that the proof of Proposition I uses of (H4), and (H3) and (FP) hold by the fixed-point item above. So that proof would give IS(A) $\vdash\ddot{\mathsf{G}}$, against the first incompleteness theorem above. ↯ □

So for the 2001 system the closed faith step fails in at least one of two places, and we do not know which. Its uniform form fails outright.

C The Fourth Step of Willard's Theorem 6

Willard's Definition 6 lets *CheckProof*(t,p,q) say that p proves t and that q proves the Σ_1 sentence (25), $\exists p\,\mathrm{Prf}(t,p)$, "$t$ is provable". Lemma 4 says that PA+ proves Willard's (26) for large k, and (27) is its instance at (k,m):

$$\forall p\,\forall n\,\exists q<n\,\{[\mathrm{Log}^{k-1}p<m\wedge\mathrm{Log}^{k}n\geq m\wedge\mathrm{Prf}(\ulcorner\mho\urcorner,p)]\supset\mathit{CheckProof}(\ulcorner\mho\urcorner,p,q)\}.$$

Willard writes "Peano Arithmetic" here. The sentence is in the U-Grounding language, so the theory meant is PA+. The proof W continues as follows. Step (28) is $\mathrm{Pair}(\ulcorner\mho\urcorner,\ulcorner\neg\mho\urcorner)$, a true bounded sentence. Steps (29) and (30) are the Group-3 axiom and its instance, $\forall p\,\forall q\,\neg[\mathrm{Prf}(\ulcorner\mho\urcorner,p)\wedge\mathrm{Prf}(\ulcorner\neg\mho\urcorner,q)]$. In step (31), no p and q satisfy *CheckProof*$(\ulcorner\mho\urcorner,p,q)$. Step (32), chained with (27), says that no short proof of $\mho$ coexists with a large n. In step (33), $\Upsilon(k,m)$ supplies the large n.

The gap. Willard derives (31) from (30), because "the definitions of CheckProof and $\mho(k-1,m)$ imply that CheckProof$(\ulcorner\mho\urcorner,p,q)$ is logically equivalent to Equation (30)'s curly bracket expression". But by Definition 6 as printed, q proves (25), and (25) is not a sentence that Pair pairs with $\mho$. It has neither the substitution nor the bound $\mathrm{Log}^{k-1}(p)<m$. A long proof of $\mho$ makes (25) true and $\neg\mho$ false. So (30) does not yield (31).

The repair (ours, a sketch). Replace (28) by the true bounded sentence $\mathrm{Pair}(\ulcorner\mho\urcorner,\ulcorner\mathrm{dual}(\mho)\urcorner)$, which Pair accepts. With (29) it gives $\forall p\,\forall q\,\neg[\mathrm{Prf}(\ulcorner\mho\urcorner,p)\wedge\mathrm{Prf}(\ulcorner\mathrm{dual}(\mho)\urcorner,q)]$ in place of (30). Then redefine *CheckProof* so that q proves $\mathrm{dual}(\mho)$. Now (31) follows from the new (30) by pure logic, since *CheckProof* is literally its bracket. It remains to show that such a q is short. Let p be a proof of $\mho=\mho(k-1,m)$ with $\mathrm{Log}^{k-1}(p)<m$. Then q can be built as a tableau list of four parts:

(i) a proof of the bounded sentence $\mathrm{Prf}(\ulcorner\mho\urcorner,\bar{p})$, which is the object that Lemma 3's sketch constructs;

(ii) a proof of the bounded sentence $\mathrm{Log}^{k-1}(\bar{p})<\bar{m}$;

(iii) a proof of the true bounded sentence $\mathrm{SUBST}(\bar{n},\ulcorner\mho\urcorner)$, which depends only on k and m;

(iv) a tableau whose root, $\mathrm{dual}(\mathrm{dual}(\mho))$, is the prenex form of $\neg\,\mathrm{dual}(\mho)$. It instantiates the two universal quantifiers at $\bar{p}$ and $\ulcorner\mho\urcorner$, removes the double negation, branches, and closes every branch against (i)–(iii). If (i)–(iii) are carried as intermediate theorems with a vacuous universal quantifier, as Willard's own bounded intermediate (28) must be, the tableau instantiates those first.

Part (iv) mentions $\bar{p}$ and $\ulcorner \mho \urcorner$ a bounded number of times. Also $\ulcorner \mho \urcorner < p$. The last root of p, dual($\mho$), is a contiguous part of p, one byte longer than $\mho$, and it begins with a symbol byte. So $\ulcorner \mho \urcorner < 64^{|\mho|} \leq \ulcorner \text{dual}(\mho) \urcorner \leq p$. Hence the logarithm of (iv)'s Gödel number is $O(\text{Log}\, p)$. The repair then needs (i)–(iii) to be short enough for Corollary 1's bound $F(p,t) < 2_a^b$ to survive. Lemma 3's sketch supplies (i). Parts (ii) and (iii) are claims of the same kind, about short proofs of bounded facts that can be checked in polynomial time, and the sketch does not address them. For the redefined *CheckProof*, (27) is no longer Lemma 4's theorem. It is this length claim. So at our level of detail, Willard's printed proof has a gap at (31), and our repair is a sketch resting on Lemma 3's sketch and on the length claims for (ii) and (iii). We therefore count Theorem 6 as a sketch, although the codification records it as full.

D Sources

Page numbers are the printed page numbers of the author copies in the repository's `nachlass/papers/`. For the 2004 chapter they are the volume's. The PDF page index differs by 2 for TR 93-10 and by 1 for 2011, 2016 and 2020. Proof statuses are those of the codification's results registry (`nachlass/codification/registry/results.md`), except where this essay revises one. Quotations were checked against the author copies. The essay's analysis note (`docs/log/2026-10-24-incompleteness-ex-machina-analysis.md`) records which were checked against page images and which against the extracted text.

TR 93-10. Quotations: pp. 6–7.

2000. Lemma 8, p. 11, full; §5, p. 15.

2001. §2, the system IS(A) and regular consistency, pp. 9–12. Theorem 4.3, p. 20, full. Lemma 7.1B and Theorems 7.2–7.3, pp. 37–40, full. Appendix B, the encoding and Eqs. 86–87, pp. 51–52. Theorems C.1–C.3, pp. 56–60: C.1–C.2 sketch, C.3 full given them.

2002a. Theorem 2.2, p. 7, sketch. Theorems 3.4–3.5, p. 11, full.

2002c. Weakest and strongest notions, p. 2.

2004. Theorem 1, p. 342, full with one step deferred. Theorem 2, p. 344, sketch.

2005. §§2–3, the language, the Levels, tableaux, $\text{IS}_D(A)$ and Eqs. 4–8, pp. 4–10; the prenex remark there is stated-only. Definitions 4–5 and Theorem 1, pp. 19–20, full. Remarks 2 and 4, p. 21; Remark 4 is stated-only. Lemmas 1–2, Theorem 2 and Eqs. 19–20, pp. 23–26, full. Theorems 3–5, pp. 27–29, full; Remark 5, p. 27, stated-only. Definitions 2, 3 and 6, pp. 11 and 33. Lemma 3, p. 33, sketch; Lemma 4 and Corollary 1, p. 33, full. Theorem 6, pp. 34–36, full as printed and a sketch here (Appendix C). Remark 8 and the Π_1^* contrast, pp. 36–37, stated-only. Appendix A, pp. 39–40.

2006a. Theorems 1–2, Pudlák's and Solovay's, p. 4, cited. Theorems 3–4, pp. 6–7, full. Remark 3, p. 24.

2007. Theorem 6, p. 29, full.

2011. Remark 6.16, p. 28.

2016. Example 3.5, p. 7; §8; Conjecture 6.6, p. 23, stated-only.

2020. Theorem 4.5, p. 12, sketch.

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